

STERLITAMAK, Russian Federation

WMO: 288250

Lat: 53.6036N

Lon: 55.9329E

Elev: 137

StdP: 99.69

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -30.4	(c) -27.2	(d) -33.7	(e) 0.2	(f) -30.4	(g) -30.3	(h) 0.2	(i) -26.9	(j) 6.4	(k) -5.9	(l) 5.7	(m) -4.9	(n) 0.4	(o) 0	(p) 0.519

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.8	(c) 32.2	(d) 19.4	(e) 30.4	(f) 18.9	(g) 28.6	(h) 18.3	(i) 21.0	(j) 28.8	(k) 20.2	(l) 27.7	(m) 19.4	(n) 26.3	(o) 2.2	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	13.6	24.0	17.6	12.8	23.3	16.8	12.1	22.6	61.4	28.7	58.5	27.7	55.8	26.6	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 5.9	(b) 5.1	(c) 4.4	(e) -33.8	(f) 34.6	(g) 3.4	(h) 1.5	(i) -36.2	(j) 35.7	(k) -38.2	(l) 36.5	(m) -40.1	(n) 37.3	(o) -42.6	(p) 38.4		
(a) 5.9	(b) 5.1	(c) 4.4	(e) -33.9	(f) 22.3	(g) 3.4	(h) 1.0	(i) -36.4	(j) 23.0	(k) -38.3	(l) 23.6	(m) -40.2	(n) 24.1	(o) -42.6	(p) 24.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.6	-13.0	-11.5	-4.7	6.2	14.4	18.5	20.4	18.8	12.7	5.2	-3.0	-10.2		
	DBStd	13.27	7.48	7.19	5.68	5.66	4.61	4.34	3.58	4.31	4.28	4.58	6.30	7.72		
	HDD10.0	3120	712	601	454	141	14	1	0	1	21	159	390	626		
	HDD18.3	5244	971	834	713	364	137	51	21	48	176	407	640	884		
	CDD10.0	1140	0	0	0	27	151	256	323	272	101	11	0	0		
	CDD18.3	224	0	0	0	1	15	56	86	62	6	0	0	0		
	CDH23.3	2428	0	0	0	13	241	628	839	633	73	0	0	0		
	CDH26.7	853	0	0	0	1	62	231	291	249	19	0	0	0		
(14) Wind	WSAvg	1.8	1.7	1.9	2.0	1.9	1.9	1.7	1.6	1.5	1.4	1.8	1.9	1.9		
(15) Precipitation	PrecAvg	548	44	35	37	31	46	56	53	56	41	51	48	50		
	PrecMax	768	87	81	89	88	157	115	161	152	75	109	93	121		
	PrecMin	350	8	1	2	0	4	3	3	11	9	17	5	5		
	PrecStd	105	23	17	23	20	32	32	34	29	21	26	27	24		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.5	3.0	10.7	25.2	30.6	33.7	33.9	34.5	29.3	20.8	10.8	2.5		
		MCWB	0.5	0.5	5.3	13.6	16.7	19.4	20.4	19.3	16.3	11.6	8.3	1.0		
	2%	DB	0.2	1.2	6.2	21.8	28.5	31.6	31.6	31.9	25.2	17.7	7.6	1.4		
		MCWB	-0.8	0.0	2.7	12.2	15.9	18.9	20.2	19.0	15.4	10.1	5.9	0.5		
	5%	DB	-1.0	0.1	4.1	18.9	26.1	29.4	29.8	29.6	22.5	14.8	5.6	0.7		
		MCWB	-2.0	-1.0	1.4	10.8	15.4	18.5	19.5	19.0	14.6	9.1	4.0	-0.2		
	10%	DB	-2.5	-1.6	2.5	15.5	23.6	27.1	28.0	26.9	19.9	12.1	3.8	-0.3		
		MCWB	-3.5	-2.7	0.6	9.2	14.0	17.6	18.8	18.0	13.4	8.3	2.4	-1.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.7	1.2	5.6	14.4	18.6	21.5	22.3	21.6	18.0	13.1	8.8	1.6		
		MCDB	1.3	2.1	10.0	23.3	26.2	29.7	30.2	29.4	26.4	18.1	10.9	2.2		
	2%	WB	-0.6	0.2	3.3	12.8	17.2	20.4	21.3	20.5	16.5	11.4	6.2	0.6		
		MCDB	0.1	1.1	5.6	20.1	25.0	28.5	28.9	28.7	22.7	15.9	7.2	1.2		
	5%	WB	-1.9	-0.9	1.9	11.3	16.2	19.4	20.5	19.6	15.4	9.9	4.1	-0.1		
		MCDB	-1.1	0.1	3.5	18.0	24.2	27.0	27.6	27.3	20.7	13.8	5.3	0.7		
	10%	WB	-3.4	-2.7	0.9	9.5	15.1	18.5	19.7	18.7	14.2	8.6	2.7	-1.3		
		MCDB	-2.5	-1.6	2.4	15.2	22.4	25.5	26.2	25.6	18.8	11.7	3.7	-0.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.3	8.8	9.0	9.9	11.9	11.3	10.8	11.0	10.2	7.6	5.2	6.6		
		MCDBR	5.4	6.6	8.8	14.7	16.3	15.2	13.9	15.1	15.0	14.1	5.6	5.1		
		MCWBR	5.1	5.8	6.2	6.9	6.1	5.3	4.6	4.9	6.4	7.7	4.4	4.8		
		MCDBR	5.2	5.7	6.9	13.7	14.5	12.9	12.0	13.2	12.5	11.6	5.2	5.2		
	5% WB	MCWBR	5.0	5.3	5.2	6.8	6.1	5.2	4.4	5.1	6.1	6.9	4.5	5.1		
(40) Clear-Sky Solar Irradiance	taub		0.299	0.325	0.377	0.423	0.406	0.399	0.413	0.435	0.384	0.355	0.326	0.300		
	taud		2.046	2.013	1.939	2.057	2.251	2.329	2.291	2.209	2.321	2.298	2.172	2.027		
	Ebn at Noon		622	737	776	798	841	851	829	782	781	714	595	530		
	Edh at Noon		74	107	144	146	127	119	122	125	99	81	66	63		
(44) All-Sky Solar Radiation	RadAvg		0.97	2.02	3.43	4.49	5.40	5.93	5.80	4.65	3.06	1.47	0.69	0.62		
	RadStd		0.11	0.18	0.34	0.46	0.51	0.68	0.52	0.47	0.37	0.31	0.15	0.12		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page